

Arun Kumar Kumari Arumugam

Software Engineer | AI Model Training & Code Evaluation | Python / Java / TypeScript

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PROFESSIONAL SUMMARY

Distinguished Engineer with 15+ years designing, implementing, and reviewing production code on high-throughput distributed systems. Fluent English technical writing; proficient in Python, Java, JavaScript, TypeScript, and SQL. Daily work evaluating AI-generated code and plans (Claude Code, GitHub Copilot, MCP agents), debugging complex workflows, and explaining trade-offs clearly. M.Tech AI/ML candidate (BITS Pilani) with hands-on instruction data, QLoRA fine-tuning, prompt evaluation harnesses, and rubric-based LLM scoring -- aligned with coding tasks, quality snippets, and reviewing model outputs for correctness, performance, and clarity.

CORE SKILLS

Programming: Python, Java, TypeScript, JavaScript, SQL; algorithms, data structures, debugging, testing. AI training: instruction datasets, SFT/QLoRA, prompt design (CoT, ReAct, JSON tools), rubric scoring. Tools: Hugging Face, PEFT, LangGraph, MCP, FastAPI, Spring Boot, NLTK, scikit-learn, Jupyter, MLflow.

AI-FOCUSED ENGINEERING & EVALUATION

Verizon -- Distinguished Engineer, Software Development

Chennai | Sep 2025 -- Present

- Lead GenAI roadmap; Agentic Universe (React, FastAPI, MCP, PostgreSQL) with human-in-the-loop agent validation.
- SRE Triaging Service (FastAPI, LangGraph): review multi-agent RCA outputs citing logs, git, Jira, and K8s context.
- Org-wide Claude Code / Copilot and MCP Agent Companion for planning, tests, and code review (Java, Python, TS).
- Enterprise architecture board; API security and quality bars for production AI and ordering platforms.

Verizon -- GenAI evaluation focus

Chennai | 2024 -- Present

- Score agent traces and LLM outputs: groundedness, citations, instruction-following, unsafe actions.
- 99.99% uptime B2B systems; mentor teams on reactive services, observability, and incident debugging.

M.TECH AI/ML PROJECTS (BITS PILANI)

Portfolio: github.com/arunkumar4189/AI-ML-Projects | ID 2024AC05045

LLM / Generative AI (Group 101)

2025 -- 2026

- Medical instruction-response JSONL; QLoRA fine-tuned GPT-2 Medium (4-bit NF4, PEFT).
- Prompt eval harness: zero/few-shot, self-critique, JSON function-calling, CoT/ToT/ReAct, strategy router.

NLP, IR, conversational AI

2025 -- 2026

- Medical QA (5 clinical docs): retrieval, Flan-T5, BART intent, verify-and-correct loop.
- Dialogue gold: 12 intents, slots/NER, multi-turn state; news/clinical IR and spelling correction.

ML, deep learning, LLMops

2025 -- 2026

- SupportSense: DistilBERT+LoRA, sentiment, summary, FAQ QA, image caption/defect (Gradio).
- Defect prediction (Random Forest); CNN; Seq2Seq summarization; bandit/Q-learning policy comparison.

SOFTWARE ENGINEERING EXPERIENCE

Verizon -- Principal Engineer / Solutions Architect

Jan 2021 -- Sep 2025

B2B ordering (Spring Reactive, GraphQL, Cassandra); bulk engine 10,000+ orders/payload; SMB Enrollment, Add-A-Line, OCR ScanID, PEGA.

Verizon -- Senior Architect / Lead Developer

Jan 2016 -- Jan 2021

Event-driven APIs, Quote-to-Order automation, Cart/Config/Payment microservices, Oracle ATG.

Tech Mahindra / Wipro

2010 -- 2016

WebLogic commerce; ATG e-commerce REST (profile, tax, shipping, promotions).

EDUCATION

- M.Tech, AI & Machine Learning -- BITS Pilani -- 2025-2027 (expected) -- 2024AC05045
- B.E., Computer Science -- Anna University, Chennai -- 2010
- Languages: English (fluent), Tamil